

Noise and Outliers

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training and Testing

Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute two medians for Mean median and unusual values. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Mean median and unusual values. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Compare summaries independently

For 1, 2, 2, 3 and 12, calculate mean and median. Then replace 12 with 2 and recalculate. Explain the change.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

6. Solve the given case

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to investigate.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 17 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Keep legitimate unusual books so the recommender can serve unusual readers.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Compute a mean

Find the mean of 4, 4, 6, 6 and 10.

My reasoning and evidence

13. W2 Compute two medians

Find medians of [9, 1, 3] and [8, 2, 4, 6].

My reasoning and evidence

14. W3 Repair an explanation

A learner deletes 18 because it is above the mean. What evidence is needed before correcting or excluding it?

My reasoning and evidence

15. W4 Compare summaries independently

For 1, 2, 2, 3 and 12, calculate mean and median. Then replace 12 with 2 and recalculate. Explain the change.

My reasoning and evidence

16. W5 Report missing evidence

A dataset is empty. Is reporting a mean of 0 justified?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Noise and Outliers

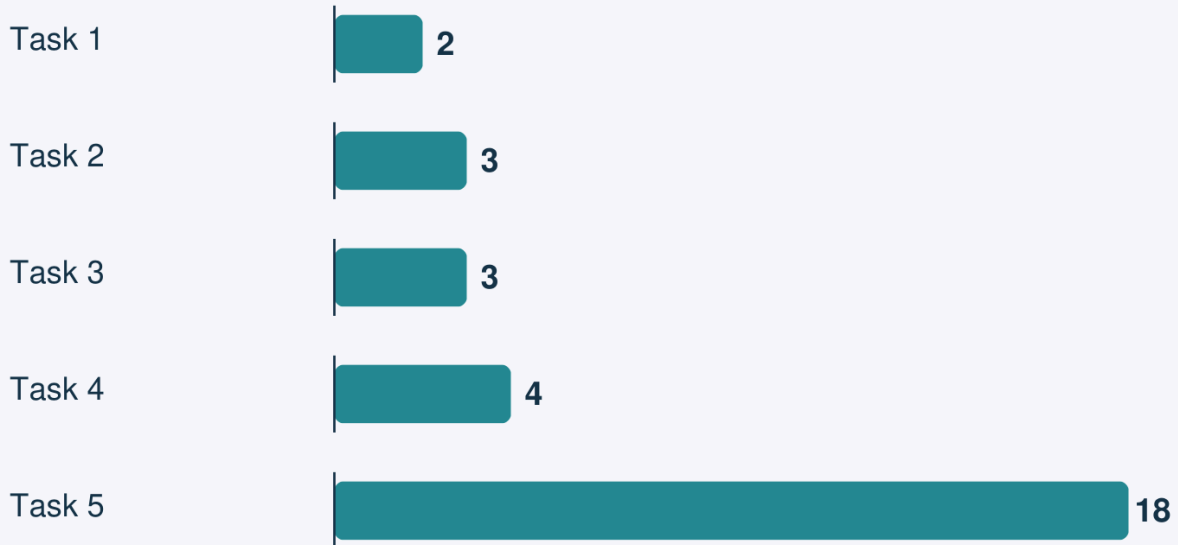
A fictional library records five book-processing times in minutes.

Sorted times: 2,3,3,4,18.

All are recorded measurements; no evidence yet shows 18 is a mistake.

Retain unusual recorded measurements

minutes



Bar length shows the supplied value; the marked baseline is zero.

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Investigate unusual recommender features and show evidence-backed repairs instead of removing inconvenient records.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Investigate unusual recommender features and show evidence-backed repairs instead of removing inconvenient records.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library records five book-processing times in minutes.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Change a value only if evidence supports a correction; preserve the raw record.

1. Median
2. Investigate 18
3. Document repair
4. Mean

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Document repair Change a value only if evidence supports a correction; preserve the raw record. The mean uses every value and is pulled upward by 18. The median identifies the middle recorded time and stays at 3. Neither statistic declares the unusual observation invalid. Deleting it simply because it changes the mean would hide potentially real slow work and change the question being answered.

Practice 2: explain the deciding evidence

Changed-case practice: An authorised log confirms 18 was a transcription error for 3. Which conclusion is supported by the supplied facts?

1. Corrected mean is 3 minutes and median is 3.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Corrected mean is 3 minutes and median is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Repair a missing building block

Review Overfitting Intuition (ai-2/16). Goal: Compare training and validation evidence for model capacity. Short practice: Does zero training error justify choosing Complex in the original record?. Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf predictions from training examples. Short practice: Why does changing colour alone not change this tree's prediction?. Return to this week and explain what you changed.

Review Training and Testing (ai-2/11). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
